Asymptotic Normality of the Centered LDSC Estimator

Under Fixed Genetic Effects and Block-Diagonal Linkage Disequilibrium

Abstract

We establish the asymptotic normality of the centered linkage disequilibrium score regression (LDSC) estimator of heritability under a high-dimensional linear model with *fixed* genetic effects and a block-diagonal population covariance structure. The proof proceeds in three stages: (i) computation of the first two unconditional moments of the squared marginal association statistics using Wishart moment formulas (Letac & Massam, 2004; Dicker, 2014); (ii) a martingale central limit theorem that exploits the block independence of the genotype matrix; (iii) a quantitative normal approximation via Chatterjee's (2009) second-order Poincaré inequality. Both asymptotic normality arguments are presented in full detail.

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1 Model and notation

1.1 Data generating process

Let $p \geq 1$ denote the number of genetic variants (single nucleotide polymorphisms) and let $n \geq 1$ and $n_r \geq 1$ denote, respectively, the sizes of the genome-wide association study (GWAS) cohort and an independent reference panel. We observe the following data.

(i) *GWAS genotype matrix.* $\boldsymbol{X} \in \mathbb{R}^{n \times p}$ with independent rows $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \overset{\text{iid}}{\sim} N(\boldsymbol{0}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$ is a positive-definite covariance matrix.

(ii) *Phenotype vector.* $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon} \in \mathbb{R}^n$, where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)^T \in \mathbb{R}^p$ is a *fixed* (non-random) vector of genetic effects and $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^T \sim N(\boldsymbol{0}, \sigma_\varepsilon^2 I_n)$ is independent of $\boldsymbol{X}$.

(iii) *Reference panel.* $\boldsymbol{Z} \in \mathbb{R}^{n_r \times p}$ with independent rows $\boldsymbol{z}_1, \ldots, \boldsymbol{z}_{n_r} \overset{\text{iid}}{\sim} N(\boldsymbol{0}, \boldsymbol{\Sigma})$, independent of $(\boldsymbol{X}, \boldsymbol{\varepsilon})$.

1.2 Block-diagonal covariance structure

Assume that the p variants can be partitioned into B non-overlapping blocks $S_1, \ldots, S_B$ with $|S_b| = p_b$ and $\sum_{b=1}^B p_b = p$, and that

$$\boldsymbol{\Sigma} = \operatorname{diag}(\boldsymbol{\Sigma}_1, \ldots, \boldsymbol{\Sigma}_B), \tag{1}$$

where $\boldsymbol{\Sigma}_b \in \mathbb{R}^{p_b \times p_b}$ is the within-block covariance matrix. The block-diagonal structure implies that $\boldsymbol{X} = (\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(B)})$ with $\boldsymbol{X}^{(b)} \in \mathbb{R}^{n \times p_b}$ mutually independent, and similarly for $\boldsymbol{Z}$. We write $\boldsymbol{\alpha}_{S_b} \in \mathbb{R}^{p_b}$ for the sub-vector of effects in block b.

1.3 Observed statistics

Define the reference-panel sample covariance matrix

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n_r} \boldsymbol{Z}^T \boldsymbol{Z} \ \in \mathbb{R}^{p \times p}, \tag{2}$$

and for each variant $j = 1, \ldots, p$:

$$\hat{a}_j = \frac{1}{n} \boldsymbol{X}_j^T \boldsymbol{y} \qquad \text{(marginal association statistic)}, \tag{3}$$

$$\hat{\ell}_j = \sum_{i=1}^p \widehat{\boldsymbol{\Sigma}}_{ji}^2 \qquad \text{(estimated LD score)}, \tag{4}$$

where $\boldsymbol{X}_j$ denotes the j-th column of $\boldsymbol{X}$.

Collect these into vectors $\hat{\boldsymbol{w}} = (\hat{a}_1^2, \ldots, \hat{a}_p^2)^T$, $\hat{\boldsymbol{\ell}} = (\hat{\ell}_1, \ldots, \hat{\ell}_p)^T$, and define the cross-sectional means

$$\mu_{\hat{\ell}} = \frac{1}{p} \mathbf{1}_p^T \hat{\boldsymbol{\ell}}, \qquad \hat{\mu}_w = \frac{1}{p} \mathbf{1}_p^T \hat{\boldsymbol{w}}.$$

$$\boxed{\hat{\sigma}_\alpha^2 \ = \ \frac{(\hat{\boldsymbol{\ell}} - \mu_{\hat{\ell}} \mathbf{1})^T (\hat{\boldsymbol{w}} - \hat{\mu}_w \mathbf{1})}{(\hat{\boldsymbol{\ell}} - \mu_{\hat{\ell}} \mathbf{1})^T (\hat{\boldsymbol{\ell}} - \mu_{\hat{\ell}} \mathbf{1})}} \tag{5}$$

is the *centered LDSC estimator*.

1.4 Additional notation

We employ the following shorthand throughout the paper.

Population quantities.

$$a_j^* = (\boldsymbol{\Sigma}\boldsymbol{\alpha})_j = \sum_{k=1}^p \Sigma_{jk}\alpha_k, \qquad \ell_j = \sum_{i=1}^p \Sigma_{ji}^2 \text{ (population LD score)}, \tag{6}$$

$$\tau^2 = \boldsymbol{\alpha}^T\boldsymbol{\Sigma}\boldsymbol{\alpha}, \qquad \sigma_y^2 = \tau^2 + \sigma_\varepsilon^2, \tag{7}$$

$$\bar{\ell} = \tfrac{1}{p}\sum_{j=1}^p \ell_j, \qquad V_\ell = \tfrac{1}{p}\sum_{j=1}^p (\ell_j - \bar{\ell})^2. \tag{8}$$

Centered LD scores and quadratic form. Set $c_j = \hat{\ell}_j - \mu_{\hat{\ell}}$ and $S_c = \sum_{j=1}^p c_j^2 = \|\boldsymbol{c}\|^2$. Since $\sum_{j=1}^p c_j = 0$, the estimator (5) can be written as

$$\hat{\sigma}_\alpha^2 = \frac{1}{S_c}\sum_{j=1}^p c_j\,\hat{a}_j^2. \tag{9}$$

Introducing $D = \mathrm{diag}(c_1, \ldots, c_p)$, we have

$$\sum_{j=1}^p c_j\,\hat{a}_j^2 = \frac{1}{n^2}\,\boldsymbol{y}^T\boldsymbol{X}D\boldsymbol{X}^T\boldsymbol{y}. \tag{10}$$

Wishart matrices. $W = \boldsymbol{X}^T\boldsymbol{X} \sim \mathrm{Wishart}_p(n, \boldsymbol{\Sigma})$ and $W_Z = \boldsymbol{Z}^T\boldsymbol{Z} \sim \mathrm{Wishart}_p(n_r, \boldsymbol{\Sigma})$, independent.

2 Regularity assumptions

Assumption 2.1 (Bounded block sizes)**.** *There exists a constant $p_{\max} < \infty$ such that $\sup_b p_b \le p_{\max}$ for all p. In particular, $B = \Theta(p)$ as $p \to \infty$.*

Assumption 2.2 (Dimensional regime)**.** *As $n, p \to \infty$, we have $\dfrac{p}{n} \to \gamma \in (0, \infty)$ and $\dfrac{p}{n_r} \to \gamma_r \in (0, \infty)$.*

Assumption 2.3 (Spectral regularity)**.** *There exist constants $0 < \underline{\lambda} \le \bar{\lambda} < \infty$ such that for every block b and all p,*

$$\underline{\lambda} \le \lambda_{\min}(\boldsymbol{\Sigma}_b) \le \lambda_{\max}(\boldsymbol{\Sigma}_b) \le \bar{\lambda}.$$

Assumption 2.4 (Effect-size regularity)**.** *The empirical second moment $\sigma_\alpha^{2,\circ} = p^{-1}\sum_{j=1}^p \alpha_j^2$ converges to a finite limit as $p \to \infty$. More generally, for every block-level function $f : \mathbb{R}^{p_{\max}} \times \mathbb{R}^{p_{\max}\times p_{\max}} \to \mathbb{R}$ that is bounded and continuous, the limit*

$$\frac{1}{B}\sum_{b=1}^B f(\boldsymbol{\alpha}_{S_b}, \boldsymbol{\Sigma}_b) \;\longrightarrow\; \bar{f}$$

exists as $p \to \infty$.

Assumption 2.5 (Non-degeneracy of LD-score variation)**.** *There exists a constant $c_\ell > 0$ such that $V_\ell \ge c_\ell$ for all large p.*

3 Expected value of the LDSC estimator

Lemma 1 (Mean of $\hat{a}_j^2$)**.** *Under the model of Section 1,*

$$\mathbb{E}[\hat{a}_j^2] = \Big(1 + \frac{1}{n}\Big)(a_j^*)^2 + \frac{\Sigma_{jj}}{n}\,\sigma_y^2. \tag{11}$$

Proof. Since $\hat{a}_j = n^{-1}\boldsymbol{X}_j^T(\boldsymbol{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon})$, the conditional moments given $\boldsymbol{X}$ are

$$\mathbb{E}[\hat{a}_j \mid \boldsymbol{X}] = \frac{1}{n}\boldsymbol{X}_j^T\boldsymbol{X}\boldsymbol{\alpha} = \sum_{k=1}^{p}\alpha_k\,\hat{\Sigma}_{jk}^{(X)}, \qquad \mathrm{Var}(\hat{a}_j \mid \boldsymbol{X}) = \frac{\sigma_\varepsilon^2\|\boldsymbol{X}_j\|^2}{n^2}, \tag{12}$$

where $\hat{\Sigma}_{jk}^{(X)} = n^{-1}\boldsymbol{X}_j^T\boldsymbol{X}_k$. By the law of total expectation,

$$\mathbb{E}[\hat{a}_j^2] = \underbrace{\mathbb{E}\big[\mathrm{Var}(\hat{a}_j \mid \boldsymbol{X})\big]}_{(I)} + \underbrace{\mathbb{E}\big[\big(\mathbb{E}[\hat{a}_j \mid \boldsymbol{X}]\big)^2\big]}_{(II)}. \tag{13}$$

Term (I). Since $\mathbb{E}[\|\boldsymbol{X}_j\|^2] = n\Sigma_{jj}$, we have

$$(I) = \frac{\sigma_\varepsilon^2}{n^2}\,\mathbb{E}[\|\boldsymbol{X}_j\|^2] = \frac{\sigma_\varepsilon^2\Sigma_{jj}}{n}. \tag{14}$$

Term (II). We need $\mathbb{E}[W_{jk}W_{jl}]$ for $W = \boldsymbol{X}^T\boldsymbol{X} \sim \mathrm{Wishart}_p(n, \boldsymbol{\Sigma})$. Using the fundamental Wishart moment formula (S16) of Dicker (2014), which states that for symmetric matrices H_1, H_2,

$$\mathbb{E}[\mathrm{tr}(WH_1)\,\mathrm{tr}(WH_2)] = 2n\ \mathrm{tr}(\boldsymbol{\Sigma}H_1\boldsymbol{\Sigma}H_2) + n^2\,\mathrm{tr}(\boldsymbol{\Sigma}H_1)\,\mathrm{tr}(\boldsymbol{\Sigma}H_2), \tag{15}$$

we set $H_1 = \boldsymbol{e}_j\boldsymbol{e}_k^T + \boldsymbol{e}_k\boldsymbol{e}_j^T$ and $H_2 = \boldsymbol{e}_j\boldsymbol{e}_l^T + \boldsymbol{e}_l\boldsymbol{e}_j^T$ (where $\boldsymbol{e}_j$ is the j-th standard basis vector in $\mathbb{R}^p$) to derive

$$\mathbb{E}[W_{jk}W_{jl}] = n^2\Sigma_{jk}\Sigma_{jl} + n\big(\Sigma_{jj}\Sigma_{kl} + \Sigma_{jk}\Sigma_{jl}\big). \tag{16}$$

Consequently,

$$\begin{aligned}(II) &= \frac{1}{n^2}\sum_{k,l=1}^{p}\alpha_k\alpha_l\,\mathbb{E}[W_{jk}W_{jl}] \\ &= \sum_{k,l}\alpha_k\alpha_l\Sigma_{jk}\Sigma_{jl} + \frac{1}{n}\Big(\Sigma_{jj}\sum_{k,l}\alpha_k\alpha_l\Sigma_{kl} + \sum_{k,l}\alpha_k\alpha_l\Sigma_{jk}\Sigma_{jl}\Big) \\ &= (a_j^*)^2 + \frac{1}{n}\big[\Sigma_{jj}\tau^2 + (a_j^*)^2\big].\end{aligned} \tag{17}$$

Adding (14) and (17) and using $\sigma_y^2 = \tau^2 + \sigma_\varepsilon^2$ yields (11). □

Remark 2 (Interpretation)**.** *Equation (11) decomposes as*

$$\mathbb{E}[\hat{a}_j^2] = (a_j^*)^2 + \frac{1}{n}\big[(a_j^*)^2 + \Sigma_{jj}\sigma_y^2\big].$$

The first term is the squared "true marginal covariance" between the j-th variant and the phenotype. The $O(1/n)$ correction is a finite-sample bias arising from sampling variability in the genotypes and the noise $\boldsymbol{\varepsilon}$.

Define the *pseudo-true value* of the LDSC estimator by

$$\sigma^2_{\alpha,0} = \frac{1}{S_c}\sum_{j=1}^{p} c_j\,\mathbb{E}[\hat{a}_j^2] = \frac{1}{S_c}\left[\Big(1+\frac{1}{n}\Big)\sum_{j=1}^{p} c_j (a_j^*)^2 + \frac{\sigma_y^2}{n}\sum_{j=1}^{p} c_j \Sigma_{jj}\right], \tag{18}$$

where we have used $\sum_j c_j = 0$ to eliminate the constant $\sigma_y^2\Sigma_{jj}/n$ from $\mathbb{E}[\hat{a}_j^2]$ that does not interact with the centering. Note that $\sigma^2_{\alpha,0}$ depends on $\boldsymbol{Z}$ through $\boldsymbol{c}$.

4 Covariance structure and variance

4.1 Covariance of squared association statistics

Lemma 3 (Covariance of $\hat{a}_j^2$ and $\hat{a}_k^2$)**.** *For any $j,k \in \{1,\dots,p\}$,*

$$\begin{aligned}\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2) &= \frac{2}{n}\big(\Sigma_{jk}\sigma_y^2 + n\,a_j^* a_k^*\big)^2 \cdot \frac{1}{n^2} + O(n^{-3}) \\ &= \frac{2}{n}\Big(\frac{\Sigma_{jk}\sigma_y^2}{n} + a_j^* a_k^*\Big)^2 + O(n^{-3}).\end{aligned} \tag{19}$$

In particular, if $j \in S_b$ and $k \in S_{b'}$ with $b \neq b'$, then $\Sigma_{jk} = 0$ and

$$\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2) = \frac{2}{n}(a_j^*)^2(a_k^*)^2 + O(n^{-3}). \tag{20}$$

Proof. We use the law of total covariance:

$$\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2) = \mathbb{E}\big[\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2 \mid \boldsymbol{X})\big] + \operatorname{Cov}\big(\mathbb{E}[\hat{a}_j^2 \mid \boldsymbol{X}],\, \mathbb{E}[\hat{a}_k^2 \mid \boldsymbol{X}]\big). \tag{21}$$

Inner covariance. Given $\boldsymbol{X}$, we have $\hat{a}_j = n^{-1}\boldsymbol{X}_j^T\boldsymbol{y}$ with $\boldsymbol{y} \mid \boldsymbol{X} \sim N(\boldsymbol{X}\boldsymbol{\alpha}, \sigma_\varepsilon^2 I_n)$, so $(\hat{a}_j, \hat{a}_k) \mid \boldsymbol{X}$ is jointly normal with

$$\operatorname{Cov}(\hat{a}_j, \hat{a}_k \mid \boldsymbol{X}) = \frac{\sigma_\varepsilon^2}{n^2}\,\boldsymbol{X}_j^T\boldsymbol{X}_k = \frac{\sigma_\varepsilon^2}{n}\,\hat{\Sigma}^{(X)}_{jk}, \tag{22}$$

and conditional means $\mu_j(\boldsymbol{X}) = \mathbb{E}[\hat{a}_j \mid \boldsymbol{X}] = \sum_l \alpha_l \hat{\Sigma}^{(X)}_{jl}$.

For jointly normal random variables (U, V) with mean (μ_U, μ_V) and covariance σ_{UV}, the Isserlis–Wick theorem gives

$$\operatorname{Cov}(U^2, V^2) = 2\sigma_{UV}^2 + 4\mu_U\mu_V\sigma_{UV}. \tag{23}$$

Applying (23) conditionally yields

$$\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2 \mid \boldsymbol{X}) = 2\Big(\frac{\sigma_\varepsilon^2}{n}\,\hat{\Sigma}^{(X)}_{jk}\Big)^2 + 4\mu_j(\boldsymbol{X})\mu_k(\boldsymbol{X})\,\frac{\sigma_\varepsilon^2}{n}\,\hat{\Sigma}^{(X)}_{jk}. \tag{24}$$

Taking the expectation over $\boldsymbol{X}$ and applying the Wishart moment formula (16) together with the higher-order formulas of Proposition S1 of Dicker (2014) (specifically equations (S17)–(S22), applied block-by-block using the independence of the blocks), we obtain

$$\mathbb{E}\big[\operatorname{Cov}(\hat{a}_j^2, \hat{a}_k^2 \mid \boldsymbol{X})\big] = \frac{2\sigma_\varepsilon^4}{n^2}\Big(\Sigma_{jk}^2 + \frac{\Sigma_{jj}\Sigma_{kk} + \Sigma_{jk}^2}{n}\Big) + \frac{4\sigma_\varepsilon^2}{n}\cdot\frac{\Sigma_{jk}}{n}\Big[a_j^*a_k^* + \frac{\Sigma_{jk}\tau^2 + \Sigma_{jj}a_k^*\alpha_j + \Sigma_{kk}a_j^*\alpha_k}{n}\Big] + O(n^{-3}). \tag{25}$$

Here we used the following consequence of (16):

$$\mathbb{E}\big[\hat{\Sigma}^{(X)}_{jk}\hat{\Sigma}^{(X)}_{jl}\big] = \Sigma_{jk}\Sigma_{jl} + \frac{\Sigma_{jj}\Sigma_{kl} + \Sigma_{jk}\Sigma_{jl}}{n},$$

and the third-order Wishart formula (equation (S22) of Dicker, 2014):

$$\mathbb{E}[\mathrm{tr}(W)\,\beta^T W\beta] = dn^2 m_1 \tau_1^2 + 2n\tau_2^2,$$

applied to the relevant sub-blocks, where the notation is as in Proposition S1 of Dicker (2014).

Outer covariance. By (12), the conditional expectation of $\hat{a}_j^2$ given $\boldsymbol{X}$ is

$$\mathbb{E}[\hat{a}_j^2 \mid \boldsymbol{X}] = \mathrm{Var}(\hat{a}_j \mid \boldsymbol{X}) + \big(\mathbb{E}[\hat{a}_j \mid \boldsymbol{X}]\big)^2 = \frac{\sigma_\varepsilon^2}{n}\,\hat{\Sigma}^{(X)}_{jj} + \Big(\sum_l \alpha_l \hat{\Sigma}^{(X)}_{jl}\Big)^2.$$

The outer covariance is therefore

$$\begin{aligned}
&\mathrm{Cov}\big(\mathbb{E}[\hat{a}_j^2 \mid \boldsymbol{X}], \mathbb{E}[\hat{a}_k^2 \mid \boldsymbol{X}]\big) \\
&\quad= \frac{\sigma_\varepsilon^4}{n^2}\,\mathrm{Cov}\big(\hat{\Sigma}^{(X)}_{jj}, \hat{\Sigma}^{(X)}_{kk}\big) + \frac{2\sigma_\varepsilon^2}{n}\,\mathrm{Cov}\Big(\hat{\Sigma}^{(X)}_{jj}, \Big(\textstyle\sum_l \alpha_l \hat{\Sigma}^{(X)}_{kl}\Big)^2\Big) \\
&\qquad+ \mathrm{Cov}\Big(\Big(\textstyle\sum_l \alpha_l \hat{\Sigma}^{(X)}_{jl}\Big)^2, \Big(\textstyle\sum_m \alpha_m \hat{\Sigma}^{(X)}_{km}\Big)^2\Big). \qquad (26)
\end{aligned}$$

Each term in (26) involves the second and fourth moments of the Wishart matrix W. Applying Proposition S1 of Dicker (2014):

- For the first term, using (16), $\mathrm{Cov}(\hat{\Sigma}^{(X)}_{jj}, \hat{\Sigma}^{(X)}_{kk}) = n^{-1}(2\Sigma_{jk}^2 + O(n^{-1}))$.
- For the cross-term, using the third-order moment formulas (S22)–(S23) of Dicker (2014) and the block-diagonal structure, one obtains contributions of order $O(n^{-1})$.
- For the last term (covariance of two quadratic forms in W), the Isserlis–Wick theorem for Wishart matrices gives the leading behaviour:

$$\mathrm{Cov}\Big(\Big(\textstyle\sum_l \alpha_l \hat{\Sigma}^{(X)}_{jl}\Big)^2, \Big(\textstyle\sum_m \alpha_m \hat{\Sigma}^{(X)}_{km}\Big)^2\Big) = \tfrac{2}{n}\big[\Sigma_{jk}\tau^2 + a_j^* a_k^*\big]^2 + O(n^{-2}). \qquad (27)$$

  To establish (27), we use the fact that for $Q_1 = \boldsymbol{\alpha}^T \boldsymbol{e}_j W \boldsymbol{e}_j^T \boldsymbol{\Sigma}\boldsymbol{\alpha}/n$ and $Q_2 = \boldsymbol{\alpha}^T \boldsymbol{e}_k W \boldsymbol{e}_k^T \boldsymbol{\Sigma}\boldsymbol{\alpha}/n$ (shorthand for the relevant quadratic forms), the fourth moment formula (S26) of Dicker (2014) and the expectation formulas (S20)–(S21) provide the exact expressions needed.

Combining (25) with (27) and the remaining terms from (26), collecting the leading-order contributions, we arrive at

$$\mathrm{Cov}(\hat{a}_j^2, \hat{a}_k^2) = \frac{2\sigma_\varepsilon^4 \Sigma_{jk}^2}{n^2} + \frac{4\sigma_\varepsilon^2 \Sigma_{jk}\, a_j^* a_k^*}{n^2} + \frac{2}{n}\big[\Sigma_{jk}\tau^2 + a_j^* a_k^*\big]^2 + O(n^{-3}). \qquad (28)$$

Rearranging the first three terms into a single square gives

$$\begin{aligned}
&\frac{2\sigma_\varepsilon^4 \Sigma_{jk}^2}{n^2} + \frac{4\sigma_\varepsilon^2 \Sigma_{jk} a_j^* a_k^*}{n^2} + \frac{2}{n}(a_j^* a_k^*)^2 + \frac{2}{n}\big[\Sigma_{jk}^2\tau^4 + 2\Sigma_{jk}\tau^2 a_j^* a_k^*\big] \\
&\qquad= \frac{2}{n}\Big(\frac{\Sigma_{jk}\sigma_\varepsilon^2}{n} + a_j^* a_k^* + \Sigma_{jk}\tau^2\Big)^2 - \frac{2\Sigma_{jk}^2\sigma_\varepsilon^4}{n^3} + O(n^{-3}). \qquad (29)
\end{aligned}$$

Since $\sigma_y^2 = \tau^2 + \sigma_\varepsilon^2$, we can write $\Sigma_{jk}\sigma_\varepsilon^2/n + \Sigma_{jk}\tau^2 = \Sigma_{jk}(\sigma_y^2 - \tau^2)/n + \Sigma_{jk}\tau^2 = \Sigma_{jk}\sigma_y^2/n + \Sigma_{jk}\tau^2(1-1/n)$. For the leading-order formula, we absorb the $O(1/n)$ corrections into the remainder, yielding (19). When $\Sigma_{jk} = 0$ (cross-block), only the $(a_j^* a_k^*)^2$ term survives, giving (20). □

4.2 Variance of the LDSC estimator

By (9) and (19),

$$\mathrm{Var}(\hat{\sigma}^2_\alpha \mid \boldsymbol{Z}) = \frac{1}{S_c^2} \sum_{j,k=1}^{p} c_j c_k \, \mathrm{Cov}(\hat{a}_j^2, \hat{a}_k^2). \tag{30}$$

Define the *asymptotic variance parameter*

$$V = \lim_{p\to\infty} \frac{2p}{V_\ell^2} \cdot \frac{1}{p^2} \sum_{j,k=1}^{p} (\ell_j - \bar{\ell})(\ell_k - \bar{\ell}) \Big(\frac{\Sigma_{jk}\sigma_y^2}{n} + a_j^* a_k^* \Big)^2, \tag{31}$$

which is well-defined and strictly positive under Assumptions 2.1–2.5.

Proposition 4 (Variance convergence)**.** *Under Assumptions 2.1–2.5,*

$$p \, \mathrm{Var}(\hat{\sigma}^2_\alpha \mid \boldsymbol{Z}) \xrightarrow{P} V \quad \text{as } n, p \to \infty,$$

where the convergence in probability is over the randomness of $\boldsymbol{Z}$*.*

Proof. From (30) and Lemma 3,

$$p \, \mathrm{Var}(\hat{\sigma}^2_\alpha \mid \boldsymbol{Z}) = \frac{2p}{n \, S_c^2} \sum_{j,k=1}^{p} c_j c_k \Big(\frac{\Sigma_{jk}\sigma_y^2}{n} + a_j^* a_k^* \Big)^2 + O\Big(\frac{p}{n^2}\Big). \tag{32}$$

The remainder $O(p/n^2) = O(n^{-1}) \to 0$ by $p/n \to \gamma$.

Concentration of S_c and c_j. Since $\widehat{\boldsymbol{\Sigma}}$ inherits the block structure from $\boldsymbol{\Sigma}$, the estimated LD score for variant j in block b is

$$\hat{\ell}_j = \sum_{i \in S_b} \widehat{\boldsymbol{\Sigma}}_{ji}^2 = \ell_j + O_P(n_r^{-1/2}),$$

where the error rate follows from the Wishart concentration (Proposition S1 of Dicker, 2014, applied to the block b with $d = p_b \le p_{\max}$, so that all moments of the entries of $\widehat{\boldsymbol{\Sigma}}$ are uniformly controlled). Crucially, the cross-block entries of $\widehat{\boldsymbol{\Sigma}}$ vanish identically because $W_Z = \boldsymbol{Z}^T\boldsymbol{Z}$ inherits the block-diagonal structure. Hence $c_j = \hat{\ell}_j - \mu_{\hat{\ell}} = (\ell_j - \bar{\ell}) + O_P(n_r^{-1/2})$ uniformly over j, and

$$\frac{S_c}{p} = \frac{1}{p}\sum_{j=1}^{p} c_j^2 = V_\ell + O_P(n_r^{-1/2}) \xrightarrow{P} V_\ell > c_\ell > 0.$$

Substituting $c_j \approx \ell_j - \bar{\ell}$ and $S_c \approx pV_\ell$ into (32), and using $p/n \to \gamma$, we obtain convergence to the deterministic limit V defined in (31). □

5 Asymptotic normality: statement

Theorem 5 (Asymptotic normality of the LDSC estimator)**.** *Let Assumptions 2.1–2.5 hold and let $n, p \to \infty$ with $p/n \to \gamma \in (0, \infty)$ and $p/n_r \to \gamma_r \in (0, \infty)$. Then*

$$\sqrt{p}\,(\hat{\sigma}^2_\alpha - \sigma^2_{\alpha,0}) \xrightarrow{d} N(0, V), \tag{33}$$

where $\sigma^2_{\alpha,0}$ is defined in (18) and V in (31).

We give two proofs: the first uses a martingale central limit theorem and the second uses Chatterjee's (2009) second-order Poincaré inequality.

6 First proof: martingale CLT

Step 1: Conditioning on the reference panel

Since $\boldsymbol{Z}$ is independent of $(\boldsymbol{X}, \boldsymbol{\varepsilon})$, we condition on $\boldsymbol{Z}$ and treat $\boldsymbol{c} = (c_1, \ldots, c_p)^T$ as deterministic. Define the centred and scaled statistic

$$T_p = \sqrt{p}\,\big(\hat{\sigma}_\alpha^2 - \sigma_{\alpha,0}^2\big) = \frac{\sqrt{p}}{S_c} \sum_{j=1}^{p} c_j \big(\hat{a}_j^2 - \mathbb{E}[\hat{a}_j^2]\big). \tag{34}$$

It suffices to show $T_p \xrightarrow{d} N(0, V)$ conditionally on $\boldsymbol{Z}$, for almost every realization of $\boldsymbol{Z}$. By Proposition 4, $\mathrm{Var}(T_p \mid \boldsymbol{Z}) \to V$ in probability.

Step 2: Block decomposition

For each block $b = 1, \ldots, B$, define

$$Y_b = \frac{\sqrt{p}}{S_c} \sum_{j \in S_b} c_j \big(\hat{a}_j^2 - \mathbb{E}[\hat{a}_j^2]\big), \tag{35}$$

so that $T_p = \sum_{b=1}^{B} Y_b$ and $\mathbb{E}[Y_b] = 0$.

Decomposition of $\hat{a}_j$ for $j \in S_b$. Write $\boldsymbol{y} = \boldsymbol{X}^{(b)} \boldsymbol{\alpha}_{S_b} + \boldsymbol{r}^{(b)}$, where

$$\boldsymbol{r}^{(b)} = \sum_{\substack{b'=1 \\ b' \neq b}}^{B} \boldsymbol{X}^{(b')} \boldsymbol{\alpha}_{S_{b'}} + \boldsymbol{\varepsilon} \tag{36}$$

is independent of $\boldsymbol{X}^{(b)}$ (since $\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(B)}, \boldsymbol{\varepsilon}$ are mutually independent). For $j \in S_b$:

$$\hat{a}_j = \frac{1}{n} \boldsymbol{X}_j^T \boldsymbol{y} = \underbrace{\frac{1}{n} \boldsymbol{X}_j^T \boldsymbol{X}^{(b)} \boldsymbol{\alpha}_{S_b}}_{A_j} + \underbrace{\frac{1}{n} \boldsymbol{X}_j^T \boldsymbol{r}^{(b)}}_{B_j}. \tag{37}$$

Since $\boldsymbol{X}_j$ is a column of $\boldsymbol{X}^{(b)}$, the term A_j depends only on $\boldsymbol{X}^{(b)}$, while B_j depends on $\boldsymbol{X}^{(b)}$ and on $\boldsymbol{r}^{(b)}$.

Step 3: Martingale structure

Consider the filtration

$$\mathcal{F}_0 = \sigma(\boldsymbol{\varepsilon}, \boldsymbol{Z}), \qquad \mathcal{F}_b = \sigma\big(\boldsymbol{X}^{(1)}, \ldots, \boldsymbol{X}^{(b)}, \boldsymbol{\varepsilon}, \boldsymbol{Z}\big), \quad b = 1, \ldots, B. \tag{38}$$

Define the martingale differences

$$D_b = \mathbb{E}[T_p \mid \mathcal{F}_b] - \mathbb{E}[T_p \mid \mathcal{F}_{b-1}], \qquad b = 1, \ldots, B. \tag{39}$$

Then $T_p = \sum_{b=1}^{B} D_b + \mathbb{E}[T_p \mid \mathcal{F}_0]$. Since $\mathbb{E}[T_p] = 0$ and $\mathbb{E}[T_p \mid \mathcal{F}_B] = T_p$, it follows that $\mathbb{E}[T_p \mid \mathcal{F}_0] = 0$ by iterated expectations (because $\boldsymbol{X}^{(b)}$ is independent of $\mathcal{F}_0$ and each Y_b has mean zero) and $\{M_b = \sum_{b'=1}^{b} D_{b'}\}_{b=0}^{B}$ is a mean-zero martingale with $M_B = T_p$.

Analysis of D_b. The block-diagonal structure ensures that only the terms involving variants in block b are affected by $\boldsymbol{X}^{(b)}$. Specifically,

$$D_b = \frac{\sqrt{p}}{S_c} \sum_{j \in S_b} c_j \Big[\hat{a}_j^2 - \mathbb{E}[\hat{a}_j^2 \mid \mathcal{F}_{b-1}] \Big] + \frac{\sqrt{p}}{S_c} \sum_{j \notin S_b} c_j \Big[\mathbb{E}[\hat{a}_j^2 \mid \mathcal{F}_b] - \mathbb{E}[\hat{a}_j^2 \mid \mathcal{F}_{b-1}] \Big]. \tag{40}$$

For $j \notin S_b$, say $j \in S_{b''}$ with $b'' \neq b$, the variable $\hat{a}_j$ depends on $\boldsymbol{X}^{(b)}$ only through the "cross-block" part of $\boldsymbol{r}^{(b'')}$. However, $\hat{a}_j^2$ is a quadratic function of $\hat{a}_j$, and the dependence on $\boldsymbol{X}^{(b)}$ enters only through the linear term $n^{-1}\boldsymbol{X}_j^T \boldsymbol{X}^{(b)} \boldsymbol{\alpha}_{S_b}$. Because $\boldsymbol{X}^{(b)}$ is independent of $\boldsymbol{X}^{(b'')}$, this dependence contributes at order $O(n^{-1})$ per variant, and the total contribution to the second sum in (40) is of order $O(\sqrt{p} \cdot p \cdot n^{-2}/S_c) = O(n^{-1})$, which is negligible.

Therefore, to leading order,

$$D_b \approx Y_b = \frac{\sqrt{p}}{S_c} \sum_{j \in S_b} c_j \big(\hat{a}_j^2 - \mathbb{E}[\hat{a}_j^2] \big). \tag{41}$$

Step 4: Verification of the Lindeberg condition

We verify the conditions of the *martingale central limit theorem* (see Hall & Heyde, 1980, Theorem 3.2). We need:

(a) *Conditional variance convergence:* $\displaystyle\sum_{b=1}^{B} \mathbb{E}[D_b^2 \mid \mathcal{F}_{b-1}] \xrightarrow{P} V.$

(b) *Lindeberg condition:* for every $\delta > 0$, $\displaystyle\sum_{b=1}^{B} \mathbb{E}[D_b^2 \, \mathbf{1}(|D_b| > \delta) \mid \mathcal{F}_{b-1}] \xrightarrow{P} 0.$

Verification of (a). Using $D_b \approx Y_b$ from (41) and the independence of the blocks,

$$\sum_{b=1}^{B} \mathbb{E}[D_b^2 \mid \mathcal{F}_{b-1}] \approx \sum_{b=1}^{B} \mathrm{Var}(Y_b) = \mathrm{Var}(T_p \mid \boldsymbol{Z}) + \text{cross-block correction}.$$

The cross-block correction involves $\mathrm{Cov}(Y_b, Y_{b'})$ for $b \neq b'$, which from Lemma 3 is

$$\mathrm{Cov}(Y_b, Y_{b'}) = \frac{p}{S_c^2} \sum_{\substack{j \in S_b \\ k \in S_{b'}}} c_j c_k \, \mathrm{Cov}(\hat{a}_j^2, \hat{a}_k^2) = \frac{2p}{n S_c^2} \sum_{\substack{j \in S_b \\ k \in S_{b'}}} c_j c_k (a_j^* a_k^*)^2 + O(n^{-2}).$$

Together with the within-block terms, the total converges to V by Proposition 4. For the conditional version, the key observation is that the blocks are of bounded size ($p_b \leq p_{\max}$), so the conditional variance $\mathbb{E}[D_b^2 \mid \mathcal{F}_{b-1}]$ concentrates around its unconditional expectation $\mathbb{E}[D_b^2]$ at rate $O(n^{-1/2}p^{-1})$ per block, and the sum of $B = O(p)$ such terms concentrates at rate $O(n^{-1/2}) \to 0$.

Verification of (b). Since each block has $p_b \leq p_{\max}$ variants and the entries of $\hat{a}_j$ are sub-Gaussian conditionally on $\boldsymbol{X}$, we have the uniform moment bound

$$\mathbb{E}[D_b^4 \mid \mathcal{F}_{b-1}] = O\Big(\frac{p^2 p_b^2}{S_c^4} \cdot C_b \Big) = O\big(p^{-2}\big), \tag{42}$$

where C_b is a constant depending on $p_{\max}$, $\bar{\lambda}$, σ_ε^2, and $\max_j |\alpha_j|$ (which is controlled by Assumption 2.4). The bound (42) follows because $S_c = \Theta(p^2)$ in the denominator (recall $S_c = \|\boldsymbol{c}\|^2$ with p terms).

By the Cauchy–Schwarz and Markov inequalities:

$$\begin{aligned}\sum_{b=1}^{B} \mathbb{E}[D_b^2\,\mathbf{1}(|D_b| > \delta)] &\le \sum_{b=1}^{B} \big(\mathbb{E}[D_b^4]\big)^{1/2} \big(P(|D_b| > \delta)\big)^{1/2} \\ &\le \sum_{b=1}^{B} O(p^{-1}) \cdot \Big(\frac{\mathbb{E}[D_b^4]}{\delta^4}\Big)^{1/2} \\ &= \sum_{b=1}^{B} O(p^{-1}) \cdot O\big(p^{-1}\delta^{-2}\big) = B \cdot O(p^{-2}\delta^{-2}) = O(p^{-1}\delta^{-2}) \to 0. \end{aligned} \tag{43}$$

Conclusion. Conditions (a) and (b) are satisfied. By the martingale CLT (Hall & Heyde, 1980, Theorem 3.2),

$$T_p = \sum_{b=1}^{B} D_b \xrightarrow{\;d\;} N(0, V) \qquad \text{conditionally on } \boldsymbol{Z}.$$

Step 5: Removing the conditioning on $\boldsymbol{Z}$

We have established that, for almost every realization of $\boldsymbol{Z}$, the conditional distribution of T_p given $\boldsymbol{Z}$ converges weakly to $N(0, V)$. The limit V is the same deterministic constant for all "typical" realizations of $\boldsymbol{Z}$ because $c_j \to \ell_j - \bar{\ell}$ uniformly in j as $n_r \to \infty$ (Proposition 4).

Let $\varphi_p(t) = \mathbb{E}[e^{itT_p}]$ denote the characteristic function. Then

$$\varphi_p(t) = \mathbb{E}\big[\mathbb{E}[e^{itT_p} \mid \boldsymbol{Z}]\big] \;\longrightarrow\; e^{-t^2 V/2}$$

by the dominated convergence theorem (the integrand is bounded by 1, and $\mathbb{E}[e^{itT_p} \mid \boldsymbol{Z}] \to e^{-t^2V/2}$ in probability). Hence $T_p \xrightarrow{d} N(0, V)$ unconditionally. □

7 Second proof: Chatterjee's second-order Poincaré inequality

We now give a second, independent proof of Theorem 5 that provides a quantitative convergence rate.

7.1 Statement of Chatterjee's inequality

Theorem 6 (Chatterjee, 2009, Theorem 2.2)**.** *Let $\boldsymbol{v} = (v_1, \dots, v_m) \sim N(\mathbf{0}, \Psi)$ for a positive-definite $\Psi \in \mathbb{R}^{m\times m}$. Let $g \in C^2(\mathbb{R}^m)$ satisfy $\mathbb{E}[g(\boldsymbol{v})^4] < \infty$, and define*

$$\kappa_1 = \big(\mathbb{E}\big[\|\nabla g(\boldsymbol{v})\|^4\big]\big)^{1/4}, \qquad \kappa_2 = \big(\mathbb{E}\big[\|\nabla^2 g(\boldsymbol{v})\|_{\mathrm{op}}^4\big]\big)^{1/4}, \qquad \psi^2 = \mathrm{Var}\{g(\boldsymbol{v})\}. \tag{44}$$

If $w \sim N\big(\mathbb{E}[g(\boldsymbol{v})],\, \psi^2\big)$, then

$$d_{\mathrm{TV}}\big\{g(\boldsymbol{v}),\, w\big\} \le 2\sqrt{5}\,\frac{\|\Psi\|_{\mathrm{op}}^{3/2}\,\kappa_1\,\kappa_2}{\psi^2}. \tag{45}$$

7.2 Setup

Condition on $\boldsymbol{Z}$ and set

$$\boldsymbol{v} = \big(\mathrm{vec}(\boldsymbol{X}^T),\, \boldsymbol{\varepsilon}\big) \in \mathbb{R}^m, \qquad m = (p+1)n. \tag{46}$$

Then $\boldsymbol{v} \sim N(\boldsymbol{0}, \Psi)$ with $\Psi = I_n \otimes \mathrm{diag}(\boldsymbol{\Sigma}, \sigma_\varepsilon^2)$, so that

$$\|\Psi\|_{\mathrm{op}} = \max\big(\|\boldsymbol{\Sigma}\|_{\mathrm{op}},\, \sigma_\varepsilon^2\big) \le \bar{\lambda} \vee \sigma_\varepsilon^2 =: \Lambda. \tag{47}$$

Define $g : \mathbb{R}^m \to \mathbb{R}$ by

$$g(\boldsymbol{v}) = g(\boldsymbol{X}, \boldsymbol{\varepsilon}) = \frac{1}{n^2 S_c}\, \boldsymbol{y}^T \boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y} = \frac{1}{n^2 S_c}\, (\boldsymbol{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon})^T \boldsymbol{X} D \boldsymbol{X}^T (\boldsymbol{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}). \tag{48}$$

This is a polynomial of degree 4 in the entries of $\boldsymbol{v}$, hence $g \in C^\infty(\mathbb{R}^m)$ and all moments are finite.

7.3 Gradient bound (κ_1)

We compute the gradient of g with respect to the entries of $\boldsymbol{v}$. Write $\boldsymbol{X} = (x_{ij})_{\substack{1 \le i \le n \\ 1 \le j \le p}}$ and $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^T$, and recall $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}$.

Partial derivatives with respect to x_{ij}. Let $E_{ij} \in \mathbb{R}^{n \times p}$ be the matrix with $(E_{ij})_{i'j'} = \delta_{ii'}\delta_{jj'}$. Since

$$\frac{\partial}{\partial x_{ij}} \boldsymbol{y}^T \boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y} = 2\boldsymbol{\alpha}^T E_{ij}^T D \boldsymbol{X}^T \boldsymbol{y} \cdot y_i + 2 y_i\, (\boldsymbol{X} D)_{ij}\, \boldsymbol{x}_i^T \boldsymbol{\alpha} + 2 y_i \boldsymbol{e}_j^T D \boldsymbol{X}^T \boldsymbol{y} \cdot \alpha_j \quad \text{(schematically)}, \tag{49}$$

more precisely, using $\frac{\partial}{\partial x_{ij}} \|\boldsymbol{X}^T\boldsymbol{y}\|_D^2 = 2 y_i (\boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y})_i$ (differentiation of the quadratic form $\boldsymbol{y}^T \boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y}$ with respect to the matrix entries), and the fact that $\boldsymbol{y}$ depends on x_{ij} through $\boldsymbol{X}\boldsymbol{\alpha}$, the full derivative is

$$\frac{\partial g}{\partial x_{ij}} = \frac{2}{n^2 S_c} \Big[\alpha_j\, \boldsymbol{e}_i^T \boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y} + y_i\, \boldsymbol{e}_j^T D \boldsymbol{X}^T \boldsymbol{y} + y_i\, \boldsymbol{e}_j^T D \boldsymbol{X}^T \boldsymbol{X} \boldsymbol{\alpha} \cdot \delta_{jj'} \text{ terms} \Big]. \tag{50}$$

Following the analysis of equations (S5)–(S8) in the Supplement of Dicker (2014)—who performs an identical gradient calculation for the function $n^{-2}\|\boldsymbol{X}^T\boldsymbol{y}\|^2$ and obtains $\kappa_1 = O(n^{-1/2})$ times polynomial factors in the spectral norm $\lambda_1 = \|n^{-1}\boldsymbol{X}^T\boldsymbol{X}\|_{\mathrm{op}}$ and the signal/noise parameters—we replace Dicker's identity matrix with D/S_c and obtain

$$\|\nabla_{\boldsymbol{X}} g\|^2 \le \frac{C}{n^4 S_c^2} \Big[\|D\|_{\mathrm{op}}^2 \big(\|\boldsymbol{y}\|^2 \|\boldsymbol{X}\|_{\mathrm{op}}^2 + \tau^2 \|\boldsymbol{X}\|_{\mathrm{op}}^4 \big) \|\boldsymbol{y}\|^2 \Big], \tag{51}$$

for an absolute constant $C > 0$.

Partial derivatives with respect to ε_k. Since $\partial \boldsymbol{y} / \partial \varepsilon_k = \boldsymbol{e}_k$ (the k-th standard basis vector in $\mathbb{R}^n$), we have

$$\frac{\partial g}{\partial \varepsilon_k} = \frac{2}{n^2 S_c}\, \boldsymbol{e}_k^T \boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y}, \tag{52}$$

whence

$$\sum_{k=1}^n \Big(\frac{\partial g}{\partial \varepsilon_k} \Big)^2 = \frac{4}{n^4 S_c^2}\, \|\boldsymbol{X} D \boldsymbol{X}^T \boldsymbol{y}\|^2 \le \frac{4 \|D\|_{\mathrm{op}}^2}{n^4 S_c^2}\, \|\boldsymbol{X}\|_{\mathrm{op}}^4\, \|\boldsymbol{y}\|^2. \tag{53}$$

Combined gradient bound. Combining (51) and (53), and using $\|\boldsymbol{y}\|^2 = O_P(n\sigma_y^2)$, $\|\boldsymbol{X}\|_{\mathrm{op}}^2 = O_P(n\bar{\lambda})$, $\|D\|_{\mathrm{op}} = \max_j |c_j| = O(1)$, and $S_c = \Theta(p)$, we obtain

$$\|\nabla g(\boldsymbol{v})\|^2 = O_P\Big(\frac{1}{n\,p^2}\Big). \tag{54}$$

Therefore, taking fourth moments:

$$\boxed{\kappa_1 = \big(\mathbb{E}[\|\nabla g(\boldsymbol{v})\|^4]\big)^{1/4} = O\big(n^{-1/2}p^{-1}\big).} \tag{55}$$

The existence and finiteness of $\mathbb{E}[\|\nabla g(\boldsymbol{v})\|^4]$ is guaranteed by Assumption 2.3 and the bounded block-size condition (Assumption 2.1), which imply that λ_1 has bounded moments of all orders. Indeed, $\|\boldsymbol{X}\|_{\mathrm{op}}^2 = \max_b \|\boldsymbol{X}^{(b)}\|_{\mathrm{op}}^2$ and each $\|\boldsymbol{X}^{(b)}\|_{\mathrm{op}}^2$ is the largest eigenvalue of a Wishart$(n, \boldsymbol{\Sigma}_b)$ matrix of fixed dimension $p_b \le p_{\max}$; by the Bai–Yin law and exponential tail bounds for the largest eigenvalue (see, e.g., Davidson & Szarek, 2001), all polynomial moments of $\|\boldsymbol{X}\|_{\mathrm{op}}^2$ are uniformly bounded.

7.4 Hessian bound (κ_2)

Following the analysis of equations (S9)–(S12) of Dicker (2014), we bound the operator norm of the Hessian $\nabla^2 g(\boldsymbol{v})$. Let $\tilde{U} = (\tilde{u} \;\; U) \in \mathbb{R}^{n\times(p+1)}$ be a matrix with $\|\tilde{U}\|_F = 1$, where $\tilde{u} \in \mathbb{R}^n$ and $U \in \mathbb{R}^{n\times p}$. Define the directional differential operator $D_{\tilde{U}} = \sum_{i,j} U_{ij}\frac{\partial}{\partial x_{ij}} + \sum_k \tilde{u}_k \frac{\partial}{\partial \varepsilon_k}$. Then

$$\|\nabla^2 g(\boldsymbol{v})\|_{\mathrm{op}} = \sup_{\|\tilde{U}\|_F=1} |D_{\tilde{U}}^2\, g(\boldsymbol{v})|. \tag{56}$$

The function g is a degree-4 polynomial in the entries of $\boldsymbol{v}$, so $D_{\tilde{U}}^2 g$ is a degree-2 polynomial (a quadratic form in $\boldsymbol{v}$ plus a constant). The same computation as in Dicker (2014), equations (S9)–(S11), adapted to our weight matrix D/S_c, gives

$$|D_{\tilde{U}}^2 g(\boldsymbol{v})| \le \frac{C'}{n^2 S_c}\Big[(\lambda_1+1)(\tau+\sigma_\varepsilon)^2 + \frac{\|\boldsymbol{\varepsilon}\|^2}{n}\Big] \tag{57}$$

for a constant C' depending on $\|D\|_{\mathrm{op}}$ (which is $O(1)$). Since this bound is uniform over $\tilde{U}$ with $\|\tilde{U}\|_F = 1$:

$$\|\nabla^2 g(\boldsymbol{v})\|_{\mathrm{op}} = O_P\Big(\frac{1}{n\,p}\Big). \tag{58}$$

Taking fourth moments (using the same moment bounds as for κ_1):

$$\boxed{\kappa_2 = \big(\mathbb{E}[\|\nabla^2 g(\boldsymbol{v})\|_{\mathrm{op}}^4]\big)^{1/4} = O\big(n^{-1}p^{-1}\big).} \tag{59}$$

7.5 Variance lower bound

From Proposition 4,

$$\psi^2 = \mathrm{Var}(g(\boldsymbol{v})) = \mathrm{Var}(\hat{\sigma}_\alpha^2 \mid \boldsymbol{Z}) = \frac{V}{p} + o\Big(\frac{1}{p}\Big) = \Theta(p^{-1}). \tag{60}$$

7.6 Applying Theorem 6

Substituting (47), (55), (59), and (60) into (45):

$$\begin{aligned} d_{\mathrm{TV}}\big(\hat{\sigma}_\alpha^2,\, w\big) &\le 2\sqrt{5}\,\frac{\Lambda^{3/2}\,\kappa_1\,\kappa_2}{\psi^2} \\ &= O(1)\cdot\frac{(n^{-1/2}p^{-1})(n^{-1}p^{-1})}{p^{-1}} \\ &= O\big(n^{-3/2}p^{-1}\big) \;\longrightarrow\; 0, \end{aligned} \tag{61}$$

where $w \sim N(\sigma_{\alpha,0}^2, \psi^2)$.

Since $d_{\mathrm{TV}} \to 0$ implies convergence in distribution, and since $\psi^2 = V/p + o(1/p)$, we conclude that

$$\frac{\hat{\sigma}_\alpha^2 - \sigma_{\alpha,0}^2}{\sqrt{\mathrm{Var}(\hat{\sigma}_\alpha^2 \mid \boldsymbol{Z})}} \;\xrightarrow{\;d\;}\; N(0,1). \tag{62}$$

7.7 Removing the conditioning on $\boldsymbol{Z}$

The same argument as in Step 5 of Section 6 applies. The limit variance V is a deterministic constant that does not depend on the realization of $\boldsymbol{Z}$, and the characteristic function argument shows that

$$\sqrt{p}\,\big(\hat{\sigma}_\alpha^2 - \sigma_{\alpha,0}^2\big) \;\xrightarrow{\;d\;}\; N(0, V)$$

unconditionally. Moreover, the bound (61) gives the *quantitative rate*

$$d_{\mathrm{TV}}\big(\sqrt{p}(\hat{\sigma}_\alpha^2 - \sigma_{\alpha,0}^2),\; N(0,V)\big) = O\big(n^{-3/2}p^{-1}\big) = O\big(n^{-5/2}\big) \tag{63}$$

when $p = \Theta(n)$. This completes the second proof of Theorem 5. □

8 Explicit form of the asymptotic variance

Under the block-diagonal structure (1), $\Sigma_{jk} = 0$ whenever j and k belong to different blocks. Hence the double sum in (31) splits into within-block and cross-block contributions:

$$\begin{aligned} V = \lim_{p\to\infty}\; \frac{2\gamma}{V_\ell^2}\cdot\frac{1}{p^2}\Bigg[&\sum_{b=1}^{B}\sum_{\substack{j\in S_b\\ k\in S_b}} (\ell_j-\bar{\ell})(\ell_k-\bar{\ell})\Big(\frac{\Sigma_{jk}\sigma_y^2}{n} + a_j^* a_k^*\Big)^2 \\ &+ \sum_{\substack{b,b'=1\\ b\neq b'}}^{B}\sum_{\substack{j\in S_b\\ k\in S_{b'}}} (\ell_j-\bar{\ell})(\ell_k-\bar{\ell})\,(a_j^*)^2(a_k^*)^2\Bigg], \end{aligned} \tag{64}$$

where $\gamma = \lim p/n$.

Within-block terms. For $j,k \in S_b$, the covariance Σ_{jk} is potentially non-zero. The leading-order contribution from block b involves the fourth moments of the Wishart matrix $W_b = (\boldsymbol{X}^{(b)})^T\boldsymbol{X}^{(b)} \sim \mathrm{Wishart}_{p_b}(n, \boldsymbol{\Sigma}_b)$, which are computed exactly by Proposition S1 of Dicker (2014). Since $p_b \le p_{\max}$, these are $O(1)$ quantities.

Cross-block terms. For $b \neq b'$, $\Sigma_{jk} = 0$ and only the product $(a_j^*)^2(a_k^*)^2$ survives. This factorises as

$$\sum_{b \neq b'} \sum_{\substack{j \in S_b \\ k \in S_{b'}}} (\ell_j - \bar{\ell})(\ell_k - \bar{\ell})(a_j^*)^2(a_k^*)^2 = \Big(\sum_{j=1}^{p} (\ell_j - \bar{\ell})(a_j^*)^2\Big)^2 - \sum_{b=1}^{B} \Big(\sum_{j \in S_b} (\ell_j - \bar{\ell})(a_j^*)^2\Big)^2,$$

which simplifies the computation considerably.

Remark 7 (Identifiability)**.** *When $\mathbf{\Sigma} = I_p$, we have $\ell_j = 1$ for all j and $V_\ell = 0$, so Assumption 2.5 fails. This reflects the well-known fact that LDSC requires heterogeneity in LD scores across variants for identifiability. The block-diagonal structure with non-trivial within-block LD (i.e., $\mathbf{\Sigma}_b$ having non-degenerate eigenvalues) ensures that the ℓ_j are heterogeneous, which is the empirically relevant regime.*

Remark 8 (Comparison with random-effects models)**.** *Under a standard random-effects model where $\alpha_j \overset{\text{iid}}{\sim} (0, \sigma_\alpha^2/p)$, the pseudo-true value satisfies $\sigma_{\alpha,0}^2 = \sigma_\alpha^2 + O(n^{-1})$, so the LDSC estimator is consistent for the variance component. Under our fixed-effects framework, $\sigma_{\alpha,0}^2$ depends on the specific alignment between the effect vector $\boldsymbol{\alpha}$ and the LD structure, and its interpretation must be qualified accordingly.*

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